



Progress towards an unassisted element identification from Laser Induced Breakdown Spectra with automatic ranking techniques inspired by text retrieval[☆]

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ABSTRACT

In this communication, we will illustrate an algorithm for automatic element identification in LIBS spectra which takes inspiration from the vector space model applied to text retrieval techniques. The vector space model prescribes that text documents and text queries are represented as vectors of weighted terms (words). Document ranking, with respect to relevance to a query, is obtained by comparing the vectors representing the documents with the vector representing the query.

In our case, we represent elements and samples as vectors of weighted peaks, obtained from their spectra. The likelihood of the presence of an element in a sample is computed by comparing the corresponding vectors of weighted peaks. The weight of a peak is proportional to its intensity and to the inverse of the number of peaks, in the database, in its wavelength neighboring.

We suppose to have a database containing the peaks of all elements we want to recognize, where each peak is represented by a wavelength and it is associated with its expected relative intensity and the corresponding element.

Detection of elements in a sample is obtained by ranking the elements according to the distance of the associated vectors from the vector representing the sample.

The application of this approach to elements identification using LIBS spectra obtained from several kinds of metallic alloys will be also illustrated. The possible extension of this technique towards an algorithm for fully automated LIBS analysis will be discussed.

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1. Introduction

In Laser-Induced Breakdown Spectroscopy (LIBS) the traditional approach for deciding about the presence of one element in the sample involves the measure of the emission intensity of the expected most intense line from the element in object. Typically, the matrix of the sample investigated is known, so that the choice of the line to be measured can be changed according to the possible interfering elements. A threshold is set for the intensity (or the signal-to-noise ratio) to decide whether the element is detected or undetected within the actual limits of detection [1]. Multi-element detection based on

multivariate analysis methods was also reported [2]. Some methods for classification of materials by comparison of spectra with spectral libraries representing different groups were also proposed [3–7].

However, LIBS applications to the recognition of materials of a priori unknown composition are constantly growing [8–10], as a consequence of the increasing availability of broadband spectrometers [11]. Unknown materials do not allow the preliminary decision on interfered lines and large amounts of spectral data must be processed. Reliable automatic algorithms able to identify elements from the characteristic spectral peaks detected in LIBS spectra are therefore needed, to support the analyst in the processing of this massive amount of data. A semiautomatic software for processing LIBS spectra has been described in [12]. However, to our knowledge no methods for fully unassisted LIBS analysis have been proposed so far.

A typical broadband LIBS spectrum may include more than thirty thousand wavelength values and corresponding intensities [13]. Although not all of those data carry significant information, in many cases the analyst has nevertheless to cope with LIBS spectra characterized by thousands of emission peaks, affected by a certain

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amount of noise. The origin of this noise can be instrumental, and then in principle reducible through a proper optimization of the experimental equipment. Among this category we could mention the electronic noise of the optical detector, which may partially or totally hinder the emission of weak peaks, the effect of a poor wavelength calibration of the spectrometer which would apparently shift the peak from its predicted spectral position, or the result of a poor spectral resolution which would broaden the spectral features of the elements in the plasma [14]. On the other hand, there are sources of “noise” arising directly from the physical properties of the laser induced plasma, such as the presence of a background emission due to continuum-continuum and continuum-bond transitions in the laser-induced plasma [15], or the physical shift and broadening of the spectral peaks due to the Stark and Doppler effects [16]. Those effects can be reduced with a careful choice of the experimental conditions but, as well as the instrumental noise, they could never be completely eliminated from the spectrum, so that the identification of the spectral peaks cannot be simply done through a comparison of the measured line emission wavelength with the theoretical one, corresponding to the observed transition. In fact, the experimental uncertainties in wavelength calibration, the finite resolution of the spectral detectors and the physical shift and broadening of the emission peaks, would eventually result in the association of several possible transitions characteristics of different parent elements to a given peak in the spectrum.

The purpose of this work is to develop a method for automatic determination of the most probable qualitative composition of the sample under study compatible with the spectral features determined by the LIBS analysis. This is obviously a pre-requisite towards the development of systems for automatic *quantitative* analysis of LIBS spectra capable of operating without the assistance of the analyst.

Assuming to have a list of experimental peaks, i.e. wavelength/intensity pairs, our algorithm compares the observed peaks to those expected from a database for each element of the periodic table. The result is a number (score) associated to each element, representing an estimate of the likelihood of the presence of the element in the sample.

The ideas presented in this paper were implemented in two versions. We produced both a Java command line implementation and a Matlab implementation.

2. Principles of the method

The basic idea of the approach that we propose is to estimate the likelihood of the presence of an element in a sample by automatically comparing the peaks of the spectra of elements and those of the sample. To obtain that we perform two steps: 1) extracting the peaks from spectral data, and 2) matching the peaks of the sample against the peaks that are supposed to be present in an element.

2.1. Peak detection

The first step in the analysis consists in the extraction of the relevant information from the spectral data. The problem of automatically determining number and position of significant emission peaks is common to many spectroscopic techniques. For obtaining the wavelength and associated intensity values corresponding to the observed emission lines, the peak can be for example identified as the occurrence of a local minimum of the second derivative of the intensity curve [17]. A threshold in the second derivative value can be defined for rejecting the instrumental noise of the spectrum background and retaining only the meaningful information. The selected pairs of wavelength-intensity values corresponding to the peaks are then stored in a vector (vector of experimental peaks).

This method is particularly powerful for locating the maxima of partially superimposed peaks; however, it is also quite sensitive

to noise. A smoothing of the LIBS spectrum with, for example, a Savitzky–Golay filter [18], would help in reducing the noise and, thus, in improving the effectiveness of the peak finding algorithm.

Since the purpose of this paper is the description of an automatic method for unassisted element detection from a list of given experimental peaks, the optimization of the peak finding algorithm will not be discussed further. In the following we will assume that a series of significant peaks is available, and we will present a method for the determination of the most probable set of elements that could be associated to such peaks. In Section 4 we discuss possible algorithms for automatic choice of the parameters used for peak finding (i.e. threshold and smoothing degree).

2.2. Use of a database

For the second step, we suppose to have a database containing the peaks of all elements we want to recognize. In the database, each peak would be represented by its wavelength, the corresponding emitting species and its intensity (relative to the other peaks of the same species). The peak database used in this work is derived from the NIST Atomic Spectra Database [19]; it contains, among other information not used for the purpose of this paper, the main emission peaks of the elements, the corresponding emitting species and an estimation of the intensity of the spectral peak. This latter parameter is important for the implementation of the method proposed; however, the theoretical peak intensities reported in the NIST database may refer to different temperature conditions compared to those typical in LIBS plasmas. Moreover, the theoretical intensity values cannot be used for comparing the intensities of the emissions of different species. One indeed could consider the possibility of recalculating the (relative) values of line intensities for each element, by measuring the plasma temperature from the LIBS spectrum and using the parameters for the transition probabilities and level degeneracies available in the literature [20]. However, it should be considered that the spectral parameters needed for such kind of calculations are available only for a limited number of lines and elements; moreover, the accuracy of the transition probabilities is extremely variable (ranging from 5 to 50% and over, for some line) and the theoretical calculation which allows to obtain the plasma temperature from the LIBS spectrum is valid only in the approximation of optically thin, homogeneous plasma in Local Thermal Equilibrium [21]. Last but not least, the determination of the plasma temperature that can be obtained from the LIBS spectrum is based on the identification of at least one element in the plasma, while for the application of the proposed method the composition of the sample is considered a priori unknown.

The theoretical intensities in the database must thus be normalized in such a way that for each species the sum of the intensities of all the peaks in the database would be equal to unit.

For each element of the periodic table, the pairs of wavelength-intensity values contained in the database are then stored in a vector (vector of the element).

2.3. Full description of the method

As anticipated before, our objective is to obtain an estimation about the presence of an element in a sample by matching the peaks that are supposed to be seen if the element is present and those that are actually detected in the sample.

In this paper we propose to automatically perform this matching taking inspirations from text retrieval techniques [22], which are widely used in text retrieval search engines, like for instance web search engines. More technically, we will build our technique using the *vector space model*, which, in case of text retrieval, prescribes that text documents and text queries are represented as vectors of weighted terms (words). The weight of a term is proportional to the importance of the term for the query or document. Document ranking

(that is, ordering documents according to their relevance to the query) is obtained computing the cosine of the angle between the query vector and the vectors representing the documents: vectors having the same direction have high relevance.

To adapt this paradigm to our context we make the following considerations. Text documents are composed of sets of terms. Elements are composed of (or, more correctly, represented by) sets of peaks. Relevance of documents to queries is computed by matching terms in documents and queries. Relevance of elements to samples is computed by matching peaks in elements and samples. In practice, the idea is that peaks, in our context, play the role of terms, elements play the role of documents, and samples play the role of queries. In text retrieval, matching between documents and queries is obtained by comparing vectors of weighted terms, here matching between elements and samples is obtained by comparing vectors of weighted peaks. Our goal here is to rank elements according to their relevance to the sample (that is, the likelihood that the elements occurs in the given sample), similarly to what in text retrieval systems is done to documents with respect to queries. Matching between samples and elements is obtained by comparing the corresponding weighted vectors, by means of the cosine of their angle.

In previous paragraphs we have introduced the notion of weighted terms and weighted peaks and we said that, in case of text retrieval systems, the weigh is an indication of the importance of a term for a document. An important point here is defining how weights are associated with peaks of elements and sample spectra. Given an element e and a peak p , the weight $w_{p,e}$ of the peak p , for the element e can be computed according to the two following intuitions: 1) the weight should be an indication of the relevance of the peak for the element e and 2) the weight should be an indication of the discriminative power of the peak.

Let's explain shortly these two intuitions. We suppose that the intensity of a peak is an indication of its relevance for the element. Thus, the first intuition suggests that the weight of a peak should be directly proportional to its intensity. In other words, if an element is supposed to have a very intense peak, the occurrence of such an intense peak in a sample spectrum might be considered as an important sign for the occurrence of the element in the sample.

The second intuition, has to do with the discriminative power (selectivity) of the peaks. Given that the identification of a peak might present some imprecision, peaks very close one to another might be confused. In other words, suppose, for simplicity, that two different elements, say e_1 and e_2 , have respectively two peaks very close one each other, say p_1 and p_2 . Now suppose that a sample spectrum has a peak p_3 very close to both p_1 and p_2 . In this case, due to the imprecision in determining the exact position of a peak, p_3 suggests that both e_1 and e_2 might be occurring in the sample. In order to control these situations, we would like to obtain a kind of measure for the selectivity of peaks. Specifically, if several peaks, corresponding to different elements, occurs in a very narrow interval (elements have high frequency in this interval), it means that these peaks are not discriminative of a specific element (the occurrence of a peak can be a marker for many elements) so their weight should be decreased to reduce false hits.

Previous intuitions can be formalized as follows. The weight $w_{p,e}$ of the peak p , for the element e is directly proportional to the peak relevance $pr_{p,e}$ of peak p in element e , and directly proportional to the selectivity sel_p of the peak p . Accordingly, we define the weight of a peak p for the element e as $w_{p,e} = pr_{p,e} \cdot sel_p$. Let us now define $pr_{p,e}$ and sel_p .

The peak relevance is computed as the ratio between the intensity of the peak p in the element e , and the sum of the intensities of all peaks of e : $pr_{p,e} = int_{p,e} / \sum_{p' \in e} int_{p',e}$, where $int_{p,e}$ is the intensity of peak p in element e .

In order to define the selectivity sel_p of the peak p , we first introduce the concept of element frequency ef_p of a peak p the in the

peak database. The element frequency of a peak p is defined as the ratio between the number of elements that have a peak very close to p and the total number of elements: $ef_p = \#\{e | p' \in range(p, \rho) \text{ and } p' \text{ is a peak of } e\} / N$, where $range(p, \rho)$ returns the set of peaks in an interval ρ of the wavelength domain, contained in the peak database, and N is the total number of elements. In our implementation of the algorithm, the interval ρ is calculated depending on the peak wavelength λ_p as $\rho = \lambda_p / K$, where K is a suitable constant. The strategy to set K is discussed in next section.

The selectivity sel_p of the peak p is small when the element frequency of p is high. Therefore, consistently to what is also typically done in text retrieval systems [22], sel_p is defined as the logarithm of the inverse of the element frequency of p , that is $sel_p = \log(1/ef_p)$, where the logarithm is used to prevent peaks with very low element frequency from having very high weight.

According to the above definition of weight, we can represent an element e as a vector of weights $[w_{p1,e}, \dots, w_{pm,e}]$, whose dimension (number of components) is equal to the total number m of peaks in the peak database and where we force $w_{p,e} = 0$, when p is not associated to the element e .

Similarly, when we have a sample s described in terms of its detected peaks, we also represent it as a vector of weighted peaks taken from the peak database. To do that, every peak p , obtained from the spectrum, is substituted by the peaks $range(p, \rho)$ contained in the database and is weighted similarly to what we described before. Also in this case we force $w_{p,s} = 0$, when p is not present in the sample s .

Let us now describe how we match elements against samples. Let v_e be the vector of weights representing the element e , and v_s the vector of weight representing the sample s . Let v_{es} be the vector obtained by setting to 0 all weights of v_s whose corresponding weight in v_e is 0. The vector v_{es} might have weights greater than zero just in correspondence of the peaks of e . The score of the element e for the sample s is obtained as the cosine of the angle between the to vectors v_e and v_{es} . Suppose, for instance (and in theory), that sample s contains e and the peaks identified in s include exactly (with the same intensity) the peaks of e . In this case the two vectors v_e and v_{es} will be identical, and the cosine will be 1. Clearly, in practice it will never be like that, given that the peaks detected in the sample will never be exactly equal to the peaks of the elements contained in the sample itself. However, the corresponding vectors will be close enough to obtain a high value of the cosine.

It is worth noting here that both, elements and samples, are represented by vectors whose dimension is equal to the total number of peaks in the database. This number is very large and computing the cosine of the angle between a sample and all elements might be very inefficient. However, consider that the vectors are very sparse. That is, vectors are zero in correspondence of most of the peaks and the relevant information is carried out by just a few peaks. Exploiting the formula $\cos(v_1, v_2) = \langle v_1, v_2 \rangle / (\|v_1\| \|v_2\|)$, provided that we normalize the two vectors, the cosine can be computed by means of the dot product of the two vectors $\langle v_1, v_2 \rangle$. In addition, the dot product can be computed just as the sum of the products of the weight of the few peaks of the sample that have weight greater than zero, with the corresponding peaks of the elements. In fact the peaks with weight zero do not give any contribution to the sum. To make this process efficient we take again inspiration from text retrieval techniques and compute the dot product incrementally [23]. More specifically, we maintain an index of all elements of the element database using as entries their peaks: given any peak p , for instance contained in a sample, we can efficiently retrieve all elements having a peak in the range $range(p, \rho)$. Therefore, all matching elements to a sample, and the corresponding ranking, are obtained by scanning the few peaks of the sample having weight greater than zero, retrieving the peaks of the database in the range $range(p, \rho)$, retrieving the elements associated with them, and computing incrementally the sum of the products between the corresponding weights to obtain the dot product for each matching element.

The result of this process is a list of elements each associated with a score, greater than zero, corresponding to the cosine of the angle between the element and the portion of the sample corresponding to the element. This list of elements can be easily ranked according to this score to have first the elements with higher score, that is the elements with higher probability to be present in the sample.

3. Examples

Before presenting some examples of the application of the proposed method to several classes of materials, we have to optimize the range (p, ρ) and to define the vector of weights for each peak in the database. It is important to point out that the optimization of the range and the assignments of weights to the peak database is done just once for all. In fact, these two aspects are independent from the sample and depend solely on the statistics/properties of the database and on the spectral resolution of the detector. They have to be recomputed just if we change the database and/or the experimental apparatus.

We found that using a range too wide ($K = 1000$) completely spoils the predictivity of the algorithm. A range corresponding to $K = 5000$ starts to produce meaningful results, but only from $K = 10,000$ on the system start to perform at its best, up to the point whereby increasing K the range becomes too small for an effective recognition of the elements ($K = 1,000,000$) and the algorithm, again, fails. It seems reasonable the hypothesis that the optimum range should be comparable with the resolution of the spectrometer used. In fact, the LIBS spectra used for this test were acquired with an Echelle spectrometer with spectral resolution $\Delta\lambda/\lambda = 5000$ [24]. In the examples following, we always used the value $K = 10,000$.

The effectiveness of the method here proposed was tested on LIBS spectra obtained from different kinds of certified materials (aluminum-, iron-, copper- and gold-based alloys) representing quite well the typical materials usually analyzed in our laboratory. The spectra were acquired with the apparatus described in [24]. The choice of parameters employed in the analysis, i.e. the threshold in the second derivative for peak finding and the range ρ , were the same in all the examples reported. No smoothing was applied to the spectra.

The output of the algorithm, as discussed in previous section, is a list of elements, ranked according to their score. The score of an element is obtained as the cosine of the angle between the associated vector and the vector representing the sample. The higher the score, the higher the probability that the element is relevant to the sample (it might be present in the sample). For the sake of clarity, we present in each example only the five resulting elements with the higher score. Elements which are in the nominal composition of the sample

Table 1

Results of the automatic algorithm for element recognition (bronze sample). The results shown in Tables from I to IV where obtained using the value $K = 10,000$.

Element	Score
Cu	0.177191
Sn	0.06011
Pb	0.058649
Ca	0.035646
Au	0.031582

are marked in bold, while surface contaminants and atmospheric elements, which are found in the LIBS spectrum with standard LIBS analysis, are marked in *italics*.

Since different ionization states of an element give rise to different spectra, neutral and singly ionized species are first ranked separately with the algorithm described above; then we obtain the element score by summing the scores of the two ionization states.

It's worth stressing at this point that the score of an element resulting from the application of the described algorithm is not correlated to its concentration in the sample. However, it gives an indication on the probability of having correctly detected its emission peaks in the LIBS spectrum and therefore of having it in the sample, at whatever concentration.

3.1. Bronze

In Fig. 1 we show the spectrum of a bronze standard, whose nominal composition is reported in the inset. The application of the algorithm presented in this paper gives the results reported in Table 1 (only the top 5 elements are reported).

As immediately manifest from the results reported in the table, at least in this simple case the algorithm correctly recognize the presence in the spectrum of the emission peaks of the major elements of the sample. Moreover, the fourth element recognized, calcium, although not present in the nominal composition of the alloy, is nevertheless present in the sample, probably in the form of surface impurity. The Ca II spectral peaks at 393.37 nm and 396.85 nm are clearly visible in the spectrum and therefore they are correctly recognized by the software (see Fig. 2).

The presence of the fifth element in the result, Au, is interesting and requires further discussion. In this case the algorithm reports the probable presence of gold emission peaks in the spectrum, with a score comparable to that of Calcium, whose peaks where indeed quite evident.

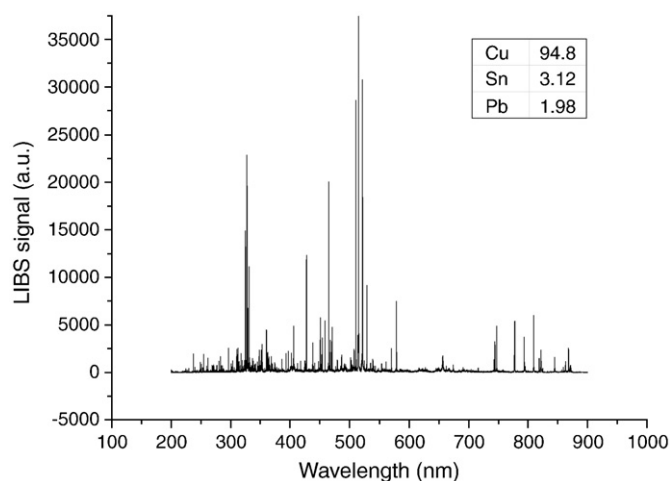


Fig. 1. LIBS spectrum of a bronze alloy; the inset shows the nominal composition (wt.%).

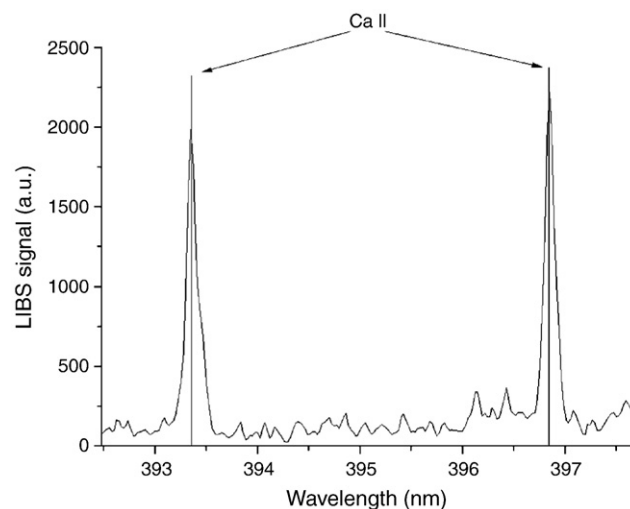


Fig. 2. Detail of the LIBS spectrum in Fig. 1, showing two Ca II emission peaks.

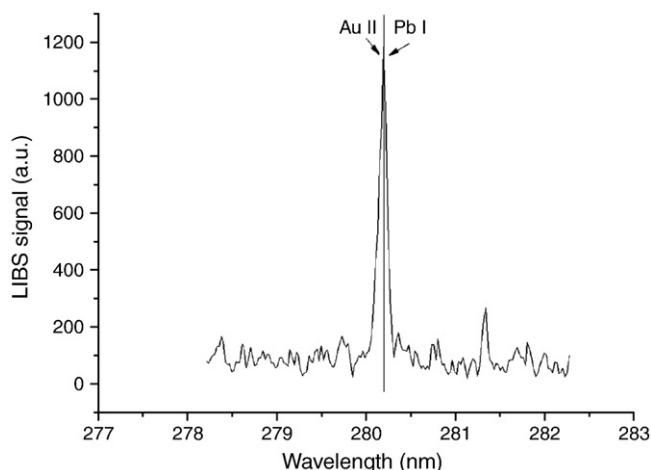


Fig. 3. Portion of the LIBS spectrum of the bronze sample, showing the superposition of the actual Pb I emission line with the theoretical wavelength associated to the most intense emission peak of Au II.

The mis-identification is due to the actual superposition of the Au II emission line at 280.202 nm with the Pb I emission line at 280.199 nm, which is in fact present in the sample (see Fig. 3). The algorithm makes no attempt of assigning the emission peak either to Pb or Au; it just gives a high score to both the elements because both of them are known to have intense emission peaks in close vicinity of the experimentally observed peak.

3.2. Aluminum alloy

In Fig. 4 we show the spectrum and the nominal composition of an aluminum alloy. The algorithm here proposed, applied to this sample, gives the results shown in Table 2.

It can be noted that in this case the algorithm correctly assigned the first five positions to elements whose emission peaks are present in the spectrum. In particular Mn, Al, Mg and Ti belong to the alloy, while nitrogen is present in the spectrum as an emission line from atmosphere. It should be noted that the algorithm does not place in the first five position of the list Fe and Zn, that are present in the sample at a concentration of about 0.2%, while for example the Ti peaks are successfully identified although the concentration of this

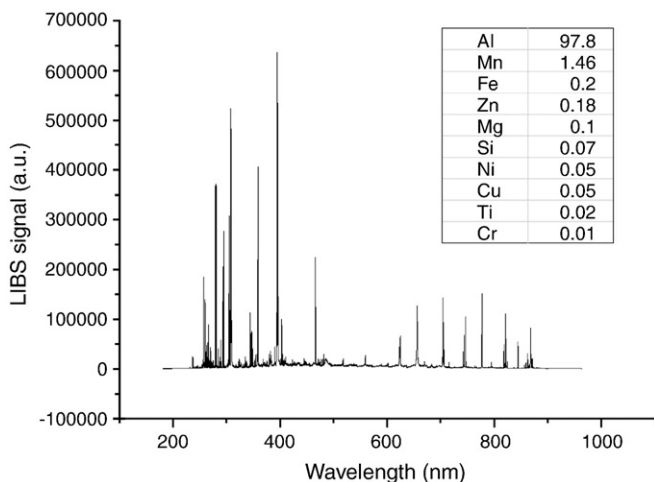


Fig. 4. LIBS spectrum of an aluminium alloy; the inset shows the nominal composition (wt.%).

Table 2

Results of the automatic algorithm for element recognition (Al-based alloy).

Element	Score
Mn	0.158302
Mg	0.089145
Al	0.085777
Ti	0.043786
N	0.036397

element is one order of magnitude less. This fact confirms the consideration that the ranking in the list of the most probable elements does not depend on the concentration of the same element in the sample (Table 3).

3.3. Steel

For exploring the possible problems related to the detection of an element, such as Fe, characterized by a great number of emission peaks, we considered the more challenging task of identifying the constituent elements of a steel sample. As usual, the LIBS spectrum and the nominal composition of the alloy are shown in Fig. 5.

LIBS spectra of steel samples are typically crowded of peaks from the major elements (Fe, Cr and Ni, in this case). The automatic algorithm, set up with the same parameters used for the analysis of the Cu- and Al-based alloys presented in Figs. 1 and 4, gives the results shown in Table 4. Note that in this case, for the sake of clarity of the discussion, the spectral region considered spans from 230 to 500 nm. In this region, in fact, we expect to find the main emission peak of the elements of the material. Therefore, the emission lines of the atmospheric elements, although present in the spectrum, would not be here considered, as well as the lines of alkaline elements (Na, for example) which have their main emission lines in the spectral region above 500 nm.

It seems that, despite the huge number of spectral peaks processed in this case (more than 1100), the algorithm manages quite well in identifying the three main components of the alloy, which hold the first three positions in the result. However, the algorithm is not able to detect the presence of Mn or Si, which are present in the sample at the level of percent. On the other hand, the mis-identification of Th and Nd is due to an effect similar to that described about the detection of Au in the bronze sample.

3.4. Precious alloy

As a final test of the algorithm, we present the results obtained on a precious alloy, whose spectrum and composition are shown in Fig. 6. The results obtained are shown in Table 4.

Also in this case the algorithm performs quite well, detecting all the components of the alloy plus magnesium, which is not a component of the alloy but whose peaks are nevertheless visible in the LIBS spectrum, presumably deriving from a surface contamination of the sample.

Table 3

Results of the automatic algorithm for element recognition (Fe-based alloy, detection range from 230 to 500 nm).

Element	Score
Fe	0.040148
Cr	0.033689
Ni	0.027187
Th	0.023087
Nd	0.019986

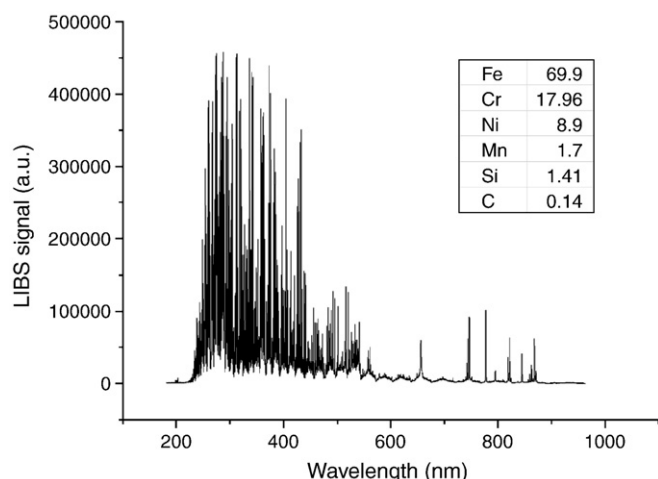


Fig. 5. LIBS spectrum of an iron-based alloy; the inset shows the nominal composition (wt.%).

4. Discussion

The results presented in the examples show that our ranking criterion usually gives a higher score to elements that are actually present in the LIBS spectrum; however, the elements present in concentrations close to their detection limits in the experimental conditions considered, are generally not well recognized; moreover, in some cases misidentifications may occur. While the main causes of misidentification are quite well understood, and are anticipated in the examples, in this section we discuss them in some detail and we indicate a roadmap for improving and validating our technique.

As illustrated in the examples, a common source of misidentification is the detection of one single line lying close to another line of an element which is actually present in the sample. In those particular cases, the algorithm used seems to suffer its inspiration by text retrieval in documents; while it is quite common that a document would host more than a single word, it's rather improbable, although not impossible, that an experimentally detected spectral peak would be the exact superposition of the emission of different elements. A test of the presence of other lines might reduce the occurrence of such kind of misidentifications. We remark that in our analysis no attempt has been done so far to assign the detected lines to the elements. However, automatic line assignment would be a natural evolution of our algorithm. In fact, the score of each element is actually calculated as a sum of contributions from each line of the experimental spectrum. This makes natural to assign a score to all elements for each detected line. A first step for consistency tests would be to assign any line to the element having the higher score corresponding to that particular line. Then it would be possible to refine the element ranking by employing the additional information of the number of assigned lines for each element. A further step may be to apply a recursive procedure, where the lines assigned to an element in the previous iteration do not contribute to the calculation of the score of the other elements in the following iteration. Similar ideas will be implemented in future

Table 4
Results of the automatic algorithm for element recognition (gold-based alloy, detection range from 230 to 500 nm).

Element	Score
Pd	0.133497
Au	0.070713
Cu	0.049258
Mg	0.046799
Ag	0.045397

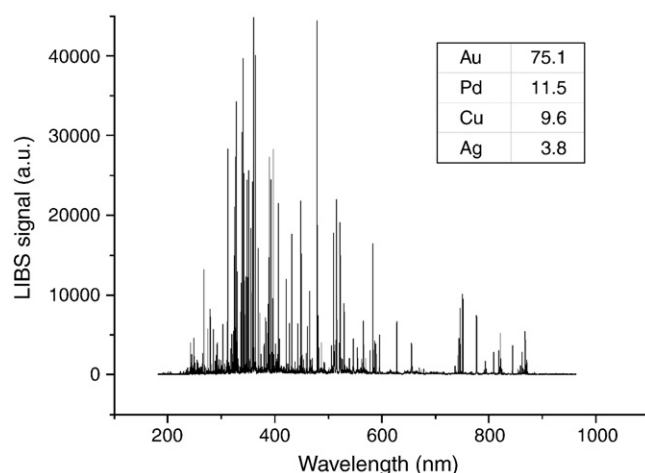


Fig. 6. LIBS spectrum of a gold-based alloy; the inset shows the nominal composition (wt.%).

works. This would be definitely useful not only for reducing mis-identifications, but also in view of the development of fully automated instrumentation for LIBS quantitative analysis based on the CF-LIBS technique [25], where the attribution of the single spectral peaks to the emitting species is clearly essential.

One of the main causes of failure for minor elements might be the poor selectivity of the method used for peak finding. In fact, the second derivative technique is very effective to separate superimposed peaks but it is less efficient to isolate weak peaks from the background fluctuations. The problem of peak detection from experimental spectra is common to all approaches to LIBS analysis, and having such step fully automated is beyond the scope of this work. However, we remark that the parameters used for peak identification are the same in all the examples reported. Improving the peak detection algorithm would be the subject of a forthcoming study.

Another limit to the effectiveness of the method is the poor reliability of the information contained in the database, as already discussed in Section 2.2. Using more suited spectral data would probably reduce the occurrence of mis-identifications. The influence of errors in the spectral data might be isolated by testing the technique on synthetic spectra; these might be obtained from the database by including some controlled perturbations, in order to simulate realistic experimental conditions. Such kind of studies would be useful to check the full consistence of the algorithm proposed independently from the spectral database; in particular, this would give a quantitative estimation of the effect of noise sources, and of the choice of parameters employed for peak identification and for ranking calculation.

Indeed, in view of a fully unassisted procedure for LIBS analysis, several possible solutions might be considered for making the choice of parameters automatic. Though the choice of parameters is quite critical, it basically depends on the chosen database and on the experimental apparatus used for the acquisition of LIBS spectra (spectral resolution, detector noise etc.); i.e. it should be optimized once for a given apparatus and database, independently on the sample under analysis; in fact, the parameters used for peak identification and for ranking calculation are the same in all the examples reported. Initial manual adjustment of parameters might be acceptable in an unassisted algorithm, provided it is independent on the sample under analysis. Nevertheless, an iterative procedure including line assignment might be helpful to automatically adjust the parameters. For example, the optimal threshold to second derivative for peak detection might be found by looking iteratively to the minimum of some function of the number of assigned peaks and of the number of

unassigned peaks. Similar considerations apply to the choice of the parameter K.

5. Conclusions

We have presented a method, inspired by text retrieval, for the automatic identification of emitting elements in LIBS spectra. The proposed technique represents a preliminary step towards a fully unassisted algorithm for LIBS analysis, to be developed in further work. The examples presented on the application of this method to bronze, aluminum, steel and precious alloys revealed that the algorithm developed performs quite well for the recognition of major components of the materials under analysis, while it does not give very exciting results for detection of elements present at concentrations close to their experimental detection limit. Future developments of this work will proceed along three main directions: improvement of the peak finding algorithm; systematic study of the influence of the different parameters, using synthetic spectra; introduction of recursive procedures for reducing the effect of spectral interference. Such studies would probably help in developing a fully automatic LIBS analytical software.

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